

DHANUSH D PRABHU

Backend & AI Systems Engineer

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SUMMARY

Backend & AI Systems Engineer building production-grade services in Python, FastAPI, and PostgreSQL, containerized with Docker. Experienced designing distributed systems and AI-powered platforms, from data pipelines to deployment. Leads engineering teams through full-stack hackathon builds, owning architecture end to end.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Bash
Backend: FastAPI, Flask, REST APIs, JWT Authentication
Frontend: React, TailwindCSS
Databases: PostgreSQL, PostGIS, SQLAlchemy
DevOps: Docker, GitHub Actions, Linux
Tools: Git, GitHub, Postman, Figma
Concepts: Data Structures & Algorithms, Database Design, Networking, Operating Systems, Distributed Systems

FEATURED PROJECT

Avana V2 — AI-Powered Safety Intelligence Platform Team Leader

Top 5 Finalist — National Level AI Hackathon, SJBIT Bengaluru • 100+ teams across Karnataka

FastAPI • PostgreSQL • PostGIS • React • TypeScript • TailwindCSS • Docker • Gemini AI

Built an AI-powered safety intelligence platform converting public incidents and community reports into actionable safety insights, deployed for live demo at a 100+ team national hackathon.

- Led architecture and technical direction for a cross-functional team, from system design through final deployment
- Designed multi-agent intelligence workflows to process and structure unstructured incident reports
- Built an automated incident extraction pipeline that parses public reports into structured, actionable data
- Implemented a geocoding pipeline to translate incident data into precise geospatial coordinates
- Engineered a risk scoring engine to quantify and rank area-level safety risk in real time
- Developed safety heatmap generation and geospatial analytics on top of PostGIS for location intelligence
- Built safe-route intelligence to recommend lower-risk paths using geospatial risk data
- Designed scalable FastAPI backend services to support concurrent multi-agent and geospatial workloads
- Built a React-based intelligence dashboard for real-time visualization of safety insights
- Coordinated team execution and delivered the final technical presentation to hackathon judges

GreenOps — Distributed Monitoring & Energy Intelligence Platform

Python • FastAPI • PostgreSQL • Docker • JWT

- Built a distributed monitoring platform with heartbeat-based agent tracking and telemetry ingestion APIs
- Implemented server-side ACTIVE / IDLE / OFFLINE state tracking from persisted heartbeat timestamps
- Designed normalized PostgreSQL schemas for high-frequency time-series telemetry writes
- Secured agent communication with JWT authentication middleware
- Containerized backend and database with Docker Compose for one-command deployment
- Built energy and uptime monitoring capability using SQL aggregation over telemetry data

Little Heartbeat Team Leader

Runner-Up — State Level Hackathon, GMIT Davangere

React • Express.js • Supabase • Gemini AI

- Led a team to build an AI-assisted pregnancy support platform with community engagement features under hackathon time constraints

LEADERSHIP

- Led engineering teams across 3+ hackathons, owning architecture decisions and technical direction end to end
- Coordinated cross-functional development efforts, keeping execution on track under strict deadlines
- Delivered technical presentations to judges and stakeholders, translating system design into clear narratives

ACHIEVEMENTS

- Top 5 Finalist — National Level AI Hackathon, SJBIT Bengaluru
- Runner-Up — State Level Hackathon, GMIT Davangere
- 1st Place — CodeNeura Inter-College Tech-a-Thon
- 1st Place — Green IT System Development Competition
- Selected for Round 2 — EY Techathon

EDUCATION

PES Institute of Advanced Management Studies Sep 2024 – May 2027
Bachelor of Computer Applications (BCA) • CGPA: 8.53 / 10